Firman Akbar Maulana
1A - Telekom

m = 10^{-3} (mili)
μ = 10^{-6} (mikro)
n = 10^{-9} (nano)

Rumus DASAR

$E = k \frac{q}{r^2}$

$k = 9 \times 10^9\ Nm^2 C^{-2}$

$F = k \frac{q_1 q_2}{r_{12}^2}$

Tugas 3 Fisika T.Telekomunikasi

1. (a) Find the electric field at point P in Fig. 2 below. (b) Repeat for $q_2 = +1$ nC.

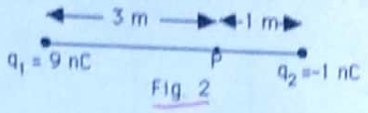


Fig. 2

2. In Fig. 3 below, $q_1 = +1.00\ \mu C$ and $q_2 = -0.0800\ \mu C$. Find the direction and magnitude of the electric field at point P.

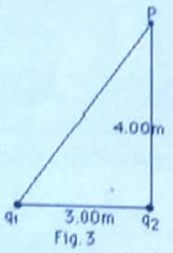


Fig. 3

3. Find (a) the electric field, direction and magnitude, at P (4.00 m, 5.00m) in Fig. 4 below due to $q_1 = +5.00 \times 10^{-6}$ C at (0, 2.00 m), $q_2 = -3.00 \times 10^{-6}$ C at (4.00 m, 0), and $q_3 = +1.6 \times 10^{-6}$ C at (0, 5.00 m) and (b) the force on $q_4 = +2.0 \times 10^{-6}$ C placed at P.

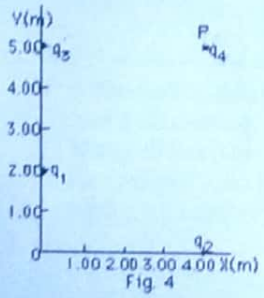


Fig. 4

agar $E = 0$

$E_1 - E_2 = 0$

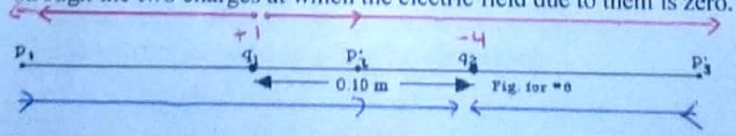
$E_1 = E_2$

$\frac{kQ_1}{r_1^2} = \frac{kQ_2}{r_2^2}$

$\left(\frac{r_1}{r_2}\right)^2 = \frac{Q_1}{Q_2}$

$\frac{r_1}{r_2} = \sqrt{\frac{Q_1}{Q_2}}$

4. Two point charges, $q_1 = +1.0 \times 10^{-6}$ C and $q_2 = -4.0 \times 10^{-6}$ C, are separated by a distance of 0.10 m. First discuss two of three points in the figure below where the electric field cannot be zero and then find the point along the line passing through the two charges at which the electric field due to them is zero.



NB: r_1 berbanding lurus dengan Q_1, sehingga semakin kecil muatan (tapi melihat arah) maka semakin mungkin muatan nol.

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5. Three infinite sheets of charge are parallel to each other, as is shown in Fig. 5 below. The sheet on the left has a uniform surface charge density $+\sigma$, the one in the middle a uniform surface charge density $-\sigma$ and the one on the right a uniform surface charge density of $+\sigma$. Find the electric field at (a) P_1, (b) P_2, (c) P_3, and (d) P_4.

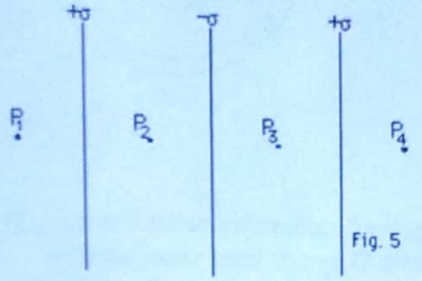
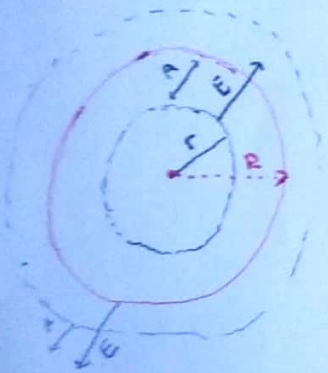


Fig. 5

6. Charge Q is distributed uniformly throughout a sphere of radius R (Fig. 6 below). Find the electric field for (a) $r \leq R$, (b) $r \geq R$. (c) Sketch a graph of E as a function of r.



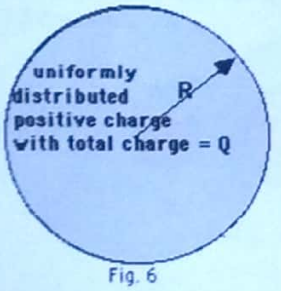


Fig. 6

$r > R$

$\oint \vec{E} \cdot d\vec{A} = EA \cdot \cos 0^\circ = \frac{Q}{\varepsilon}$

$E \cdot 4\pi r^2 \cdot 1 = \frac{Q}{\varepsilon}$

$E = \frac{Q}{4\pi\varepsilon r^2}$

$r < R$

$\oint \vec{E} \cdot d\vec{A} = EA \cos 0^\circ = \frac{q}{\varepsilon}$

$E \cdot 4\pi r^2 \cdot 1 = \frac{\left(\frac{r}{R}\right)^3 \cdot Q}{\varepsilon}$

$E = \frac{Qr}{4\pi\varepsilon R^3}$

$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$

$q = \rho \cdot \frac{4}{3}\pi r^3 = \left(\frac{r}{R}\right)^3 Q$

7. A conducting sphere of charge +Q and radius a, is concentric with a conducting, spherical shell that has a net charge of 0. The shell has inner radius b and outer radius c. (a) Where are the charges? Find the electric field for (b) $r \leq a$, (c) $a \leq r \leq b$, (d) $b \leq r \leq c$ and (e) $r \geq c$.

8. A point charge $q_i = 8.85\ \mu C$ is at the center of two concentric conducting shells as shown in Fig. 7 below. The net charge on the inner shell is $-17.70\ \mu C$ and the net charge on the outer shell is $+8.85\ \mu C$. Find (a) the inner charge q_a and the outer charge q_b on the inner shell and (b) the inner charge q_c and the outer charge q_d on the outer shell. (c) Sketch a graph of the net flux through a Gaussian sphere centered on the point charge as a function of the distance r from the point charge.

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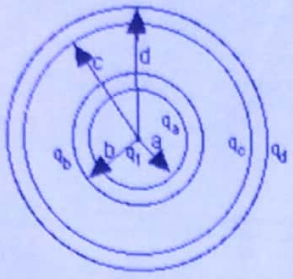


Fig. 7

*9. Figure 8 below represents four spheres of different radii R, but with the same total charge Q distributed uniformly through the spheres. Point P in the figures is the same distance from the center of each sphere. (a) Rank the spheres according to their volume charge density ρ, greatest first. (b) Rank the spheres according to the magnitude of the electric field they produce at point P, greatest first.

*Taken from Halliday./Resnick/ Walker, Fundamentals of Physics, Sixth Edition, Wiley

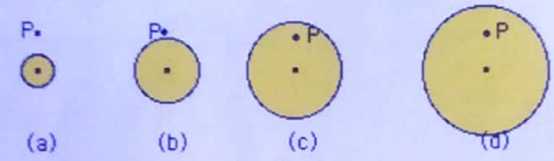


Fig. 8

10. In Fig below, there are three nonconducting rods, one circular and two straight. Each has a uniform charge of magnitude Q along its top half and along its bottom half. Find the direction of the resultant electric field at point P for each of the rods.

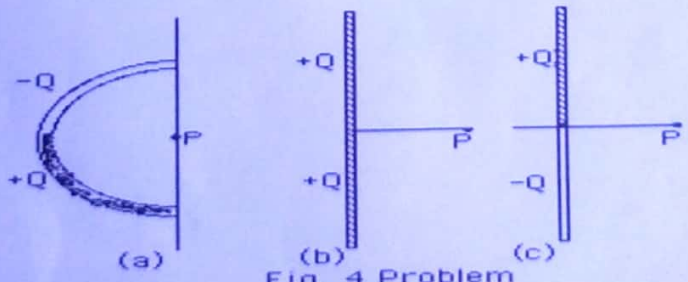


Fig. 4 Problem

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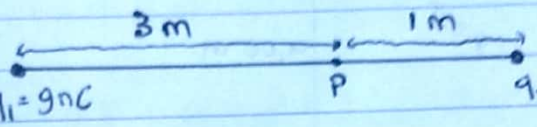
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Tugas 3 fisika

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1) a) Find the electric field at point P in fig 2 below.
b) Repeat for $q_2 = +1\ nC$



Answer :

(a) $E_1 = k \cdot \frac{q_1}{r_1^2} = \frac{9 \cdot 10^9\ Nm^2 \cdot C^{-2} \cdot 9 \cdot 10^{-9}\ C}{(3m)^2}$

$= \frac{81\ N m^2 \cdot C^{-1}}{9 m^2}$

$= 9\ N/C$

$E_2 = k \cdot \frac{q_2}{r_2^2} = \frac{9 \cdot 10^9\ Nm^2 \cdot C^{-2} \cdot (-10^{-9}\ C)}{1 m^2}$

$= \frac{-9\ N m^2 \cdot C^{-1}}{m^2}$

$= -9\ N/C$

Electric field in point p → $E = E_1 + E_2$

$= 9 + 9 = 18\ N/C$

(b) $E_1 = k \cdot \frac{q_1}{r_1^2} = \frac{9 \cdot 10^9\ Nm^2 \cdot C^{-2} \cdot 9 \cdot 10^{-9}\ C}{(3m)^2}$

$= \frac{81\ Nm^2 \cdot C^{-1}}{9 m^2}$

$= 9\ N/C$

$E_2 = k \cdot \frac{q_2}{r_2^2} = \frac{9 \cdot 10^9\ Nm^2 \cdot C^{-2} \cdot 10^{-9}\ C}{m^2}$

$= \frac{9\ Nm^2 \cdot C^{-1}}{m^2} = 9\ N/C$

Electric field in point p → $E = E_1 - E_2$

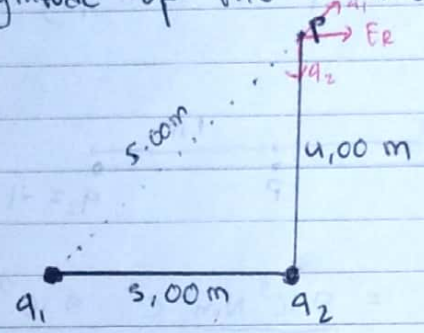
$= 9 - 9$

$= 0\ N/C$

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2) In Fig 3 below, $q_1 = +1.00\,\mu C$ and $q_2 = -0.0800\,\mu C$. Find the direction and magnitude of the electric field at point P



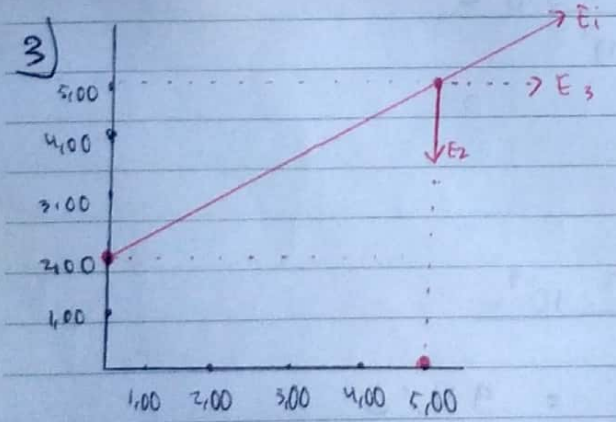
Answer :

$$E_1 = k\frac{q_1}{r_1^2} = \frac{9\cdot10^9\cdot10^{-6}}{5^2} = \frac{9\cdot10^3}{25} = 360\ N/C$$

$$E_2 = \frac{k\,q_2}{r_2^2} = \frac{9\cdot10^9\cdot8\cdot10^{-9}}{4^2} = \frac{72\cdot10}{16} = 45\ N/C$$

$$E_R = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos\theta}$$
$$= \sqrt{360^2 + 45^2 + 2\cdot360\cdot45\cos 135^\circ}$$
$$= \sqrt{129600 + 2025 + 32400\cdot -4/5}$$
$$= \sqrt{129600 + 2025 - 25920}$$
$$= \sqrt{105705} = 325{,}123\ N/S$$

3)



$q_1 = +5.00 \times 10^{-6}\,C$
$q_2 = -3.00 \times 10^{-6}\,C$
$q_3 = +1,6 \times 10^{-6}\,C$
$q_4 = +2.0 \times 10^{-6}\,C$

JOYKO 36 Lines, 6 mm

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a) * $E_1 = \frac{k \cdot Q_1}{r_{1P}^2} = 9 \cdot 10^9 \cdot \frac{5 \cdot 10^{-6}}{5^2} = 18 \times 10^2$ N/C

$E_{1x} = E_1 \cdot \cos\theta = 18 \times 10^2 \times \frac{4}{5} = 1440$ N/C

$E_{1y} = E_1 \cdot \sin\theta = 18 \times 10^2 \times \frac{3}{5} = 1080$ N/C

* $E_2 = \frac{k \cdot Q_2}{r_2 P^2} = 9 \times 10^9 \cdot \frac{3 \times 10^{-6}}{5^2} = 1080$ N/C

* $E_3 = \frac{k \cdot Q_3}{r_3 P^2} = 9 \times 10^9 \cdot \frac{1,6 \times 10^{-6}}{4^2} = 900$ N/C

$\Sigma Ex = Ex + E_3 = 1440 + 900 = 2340$ N/C

$\Sigma Ey = E_{1y} + (-E_2) = 1080 - 1080 = 0$ N/C

$E_R = \Sigma Ex = \boxed{2340 \text{ N/C}}$

b) F_4 for Q_4

$F_{4.1} = \frac{k \cdot Q_4 \cdot Q_1}{r_{4.1}^2} = \frac{9 \cdot 10^9 \times 2 \cdot 10^{-6} \times 5 \cdot 10^{-6}}{25} = 3,6 \times 10^{-3}$ N

$F_{4.2} = \frac{k \cdot Q_4 \cdot Q_2}{r_{4.2}^2} = \frac{9 \cdot 10^9 \times 2 \cdot 10^{-6} \times 3 \times 10^{-6}}{25} = 2,16 \times 10^{-3}$ N

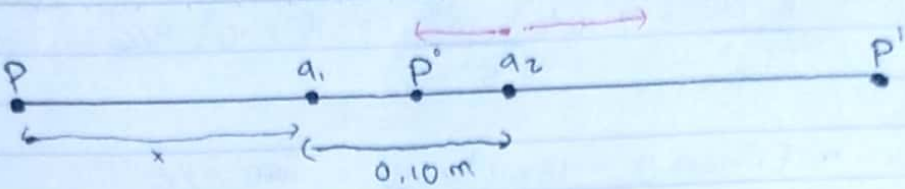
$F_{4.3} = \frac{k \cdot Q_4 \cdot Q_3}{r_{4.3}^2} = \frac{9 \cdot 10^9 \times 2 \cdot 10^{-6} \times 16 \times 10^{-7}}{16} = 1,8 \times 10^{-3}$ N

$F_R = 3,6 \times 10^{-3} + 2,16 \times 10^{-3} + 1,8 \times 10^{-3}$

$= \boxed{7,56 \times 10^{-3} \text{ N}}$

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4.



$q_1 = +1 \cdot 10^{-6}\ C$

$q_2 = -4 \cdot 10^{-6}\ C$

* = 0

karena muatan q_2 negatif maka medan listrik diantara P° dan P' adalah saling meniadakan, artinya = 0

* ≠ 0

di P → P'

$E_1 = E_2$

$$\frac{9 \cdot 10^9 \cdot 10^{-6}}{x^2} = \frac{9 \cdot 10^9 \cdot -4 \cdot 10^{-6}}{(x+0,10)^2}$$

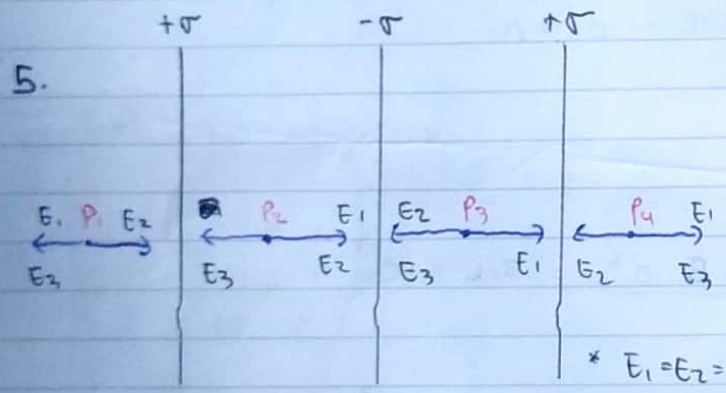
$$\sqrt{\frac{1}{x^2}} = \sqrt{\frac{4}{(x+0,10)^2}}$$

$$\frac{1}{x} = \frac{2}{x+0,10}$$

$$x + 0,10 = 2x$$

$$x = 0,10$$

5.



* $E_1 = E_2 = E_3 = E_4 = E$

dit : medan listrik di titik P

$$\oint \vec{E} \cdot \vec{dA} = \frac{Q}{\varepsilon}$$

$$2EA = \frac{Q}{\varepsilon}$$

$$E = \frac{Q}{2\varepsilon A} = \boxed{\frac{\sigma}{2\varepsilon}}$$

a.) $P_1 \cdot E_R = E_2 - E_1 - E_3$

$$= \boxed{-E = \frac{-\sigma}{2\varepsilon}}$$

b.) $P_2 = E_R = E_1 + E_2 - E_3$

$$= \boxed{E = \frac{\sigma}{2\varepsilon}}$$

c.) $P_3 = E_R = E_1 - E_2 - E_3$

$$= \boxed{-E = \frac{-\sigma}{2\varepsilon}}$$

d) $P_4 = E_R = E_1 + E_3 - E_2$

$$= \boxed{E = \frac{\sigma}{2\varepsilon}}$$

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